Analysis of Efficiency of Dark Photon Decay to Lepton Jets

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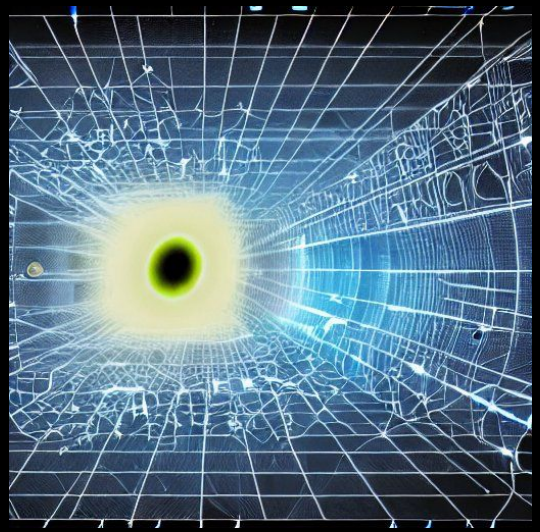
Overview

- Methodology for Efficiency Analysis
- Signal Analysis Results
- Background Analysis Results
- Conclusions

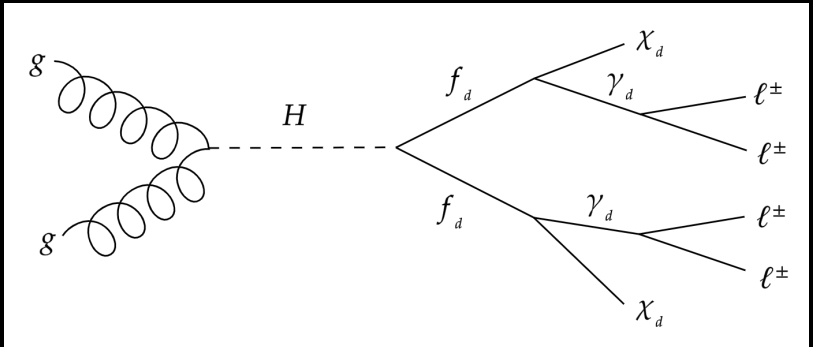


Relevance of Dark Photons

- Dark photons serve as useful indicators to study dark matter and probe new physics beyond the Standard Model.
- Lepton jets produced from dark photon decays can provide insights into the flavor content of the jets, which can be used to test theoretical models of particle interactions.
- Analysis of the varying-flavor pairs in lepton jets can offer evidence for possible connections and search for new particles or phenomena beyond the Standard Model.

Introduction to Dark Photons & Importance

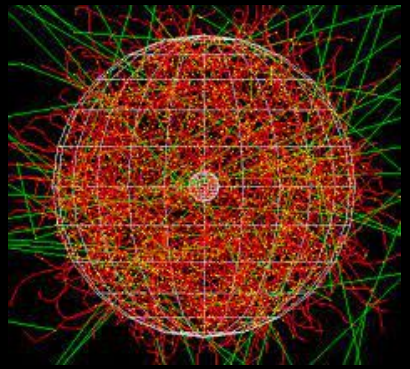
- Dark photons are particles that can explain phenomena beyond the Standard Model of particle physics, specifically in the context of dark matter



- Dark photons can interact with Standard Model particles through a process called kinetic mixing

- Kinetic mixing allows dark photons to decay into Standard Model leptons, resulting in the production of lepton jets

Methodology for Efficiency Analysis



- Use of data-driven estimates to calculate lepton jet efficiency, as Monte Carlo simulations may not be reliable for rare cases
- Original focus on lepton jets containing both muons and electrons to compare flavor combinations
- Count the number of electrons and muons in each lepton jet and compare the counts to determine if the jet is fake

Background Results and Signal Analysis

- Maintained 15-18% fake ratio in background results
- In previous sample, similar results to the signal analysis, indicating consistent fake ratios, indicating insufficient differentiability
- Previously, only jets with exactly one electron and one muon were considered invalid

- Maintained ~1% fake ratio in signal results
- New implication that fake ratios *do not* stay consistent between signal background measurements, indicating better differentiation
- Old observations with signal and background fake ratio similarity no longer valid
- Now invalid signatures are whenever both muons and electrons > 0

Old Results **(electrons and muons both exactly = 1)**	**DRELL-YAN (BACKGROUND)**	**SIGNAL**
FAKE JETS	727	45
TOTAL RECONSTRUCTED JETS	3479	278
RATIO	20.89%	16.18%

cont. with New Results

.5 CONE SIZE (electrons and muons both > 0)	DRELL-YAN (BACKGROUND)	SIGNAL
FAKE JETS	413	4
TOTAL RECONSTRUCTED JETS	2244	285
RATIO	18.4%	1.4%

.3 CONE SIZE (electrons and muons both > 0)	DRELL-YAN (BACKGROUND)	SIGNAL
FAKE JETS	306	4
TOTAL RECONSTRUCTED JETS	1948	282
RATIO	15.7%	1.41%

.1 CONE SIZE (electrons and muons both > 0)	DRELL-YAN (BACKGROUND)	SIGNAL
FAKE JETS	217	3
TOTAL RECONSTRUCTED JETS	1490	273
RATIO	14.6%	1.1%

Conclusions

- Previous results from signal and background analyses indicate consistent fake ratios in lepton jets between signal and background
- New parameters yield differentiable results from signal and background analyses indicate inconsistent fake ratios in lepton jets.
- However, due to the low overall background representation of ~20%, these conclusions support that multi-flavor lepton jets are not a good candidate for a control sample or fake estimate
- Further lepton jet type efficiency analysis helps determine background contribution and improves identification of dark photon signatures.

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